Matthew Nuñez

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Education

Massachusetts Institute of Technology — Cambridge, MA
Bachelor of Science in Physics — *Aug. 2022 – May 2026*

Research

Examining Ionization of the First Galaxies with JWST Spectroscopy — August 2025 – May 2026
Undergraduate Researcher — *MIT Kavli Institute*

- Supervisors: Professor Anna-Christina Eilers and Dr. Yuzo Ishikawa
- Performed spectral modeling on JWST data to identity and characterize key spectral features of early galaxies with redshifts 6-7
- Constructed models of emission lines and local continuum, then fit models to data from EIGER and CAPERS catalogues
- Quantified key line ratios and connected them to galaxy features such as ionization history, density, temperature, and radial velocity

Enhancing Source Detection in JWST Data with Machine Learning — February 2025 – August 2025
Undergraduate Researcher — *MIT Kavli Institute*

- Supervisors: Professor Anna-Christina Eilers and Dr. Gene Leung
- Trained, implemented, and evaluated a machine-learning algorithm to differentiate between falsely detected sources and true sources in data from JWST's Mid-Infrared Instrument
- Performed dimensionality reduction with principle component analysis (PCA) and T-distributed stochastic neighbor embedding (t-SNE) methods
- Created a machine-learning algorithm that scored images on their likelihood of being a real source, trained on a catalogue of hand-selected real and spurious sources

Visiting Researcher at Teide Observatory — January 2025
Visiting Undergraduate Researcher — *Instituto de Astrofísica de Canarias*

- Supervisors: Dr. Michael Person, Dr. Artem Burdanov, and Jensen Lawrence
- Conducted research at Teide Observatory as one of nine undergraduates selected as visiting researchers
- Operated the IAC80, Artemis, and TCS telescopes to collect data on various astrophysical sources, including NEO Alinda
- Used AstroImageJ to reduce raw telescope images from telescopes to science images ready for analysis
- Constructed light curves of Koronis family asteroids (to support research on asteroid spin vector alignment by Dr. Stephen Slivan) and Uranian satellites (to support research for MIT master's degree student Abigail Colclasure) with Python and AstroImageJ
- Presented results to the scientific community at the Instituto de Astrofísica de Canarias

High Altitude Balloon Space Imaging Experiment — September 2023 – January 2024
Undergraduate Researcher — *MIT AeroAstro*

- Supervisors: Professor Olivier de Weck and Christopher Tommila
- Designed a weather balloon for stratosphere-based imaging of the Sun, moon, and satellites
- Researched and implemented active and passive stabilization measures to steady the payload and improve imaging
- Used Fusion360 to design an FAA-compliant payload that housed and protected electronics while allowing unobstructed camera view out from the payload

Design of Cryogenic Loop System January 2023 – August 2023
Undergraduate Researcher *MIT Nuclear Science and Engineering*

- Supervisors: Professor Matteo Bucci and Marco Graffiedi
- Designed a cryogenic closed-loop system to efficiently study heat transfer characteristics of liquid hydrogen and liquid neon
- Used SolidWorks to design a test section that maintained a stable flow and pressure of cryoliquid while providing optical access to the test section
- Manufactured a prototype test section out of borosilicate glass and aluminum

Publications and Presentations

Publications

- Brothers, Timothy C. et al. (including **Nunez, Matthew**), 2025. "Lightcurve and Rotation Period of Near-Earth Asteroid 887 Alinda During the 2025 Close Approach". *Minor Planet Bulletin*, 52, 211-213.

Presentations

- **Nunez, M.** "Rotational Light Curves of Uranian Satellites", *MIT Astronomy Field Camp*, Instituto de Astrofísica de Canarias, 27 January 2025

Projects

Exploring Galactic Age-Color Relationships | MIT Principles of Remote Sensing Spring 2025

- Operated telescopes at MIT's Wallace Astrophysical Observatory to image various galaxies
- Created an analysis pipeline in Python to compare the magnitudes of different optical bands across each galaxy
- Created visuals to illustrate the different color indices of each galaxy, revealing the differences in their star-forming regions
- Presented findings to MIT undergraduates and graduate students

Photometric Analysis of AV Pegasi | MIT Optical Astronomy Fall 2024

- Operated telescopes at MIT's Wallace Astrophysical Observatory to collect photometric data on the RR Lyrae variable star AV Pegasi
- Reduced and analyzed images in Python to track the motion and changing magnitude of AV Pegasi and comparison stars, yielding a light curve
- Used known relations to calculate the distance to AV Pegasi and cross-checked value with that found from GAIA parallax

Spectroscopic Analyses of Metal-Poor Stars | MIT Stellar Archaeology Fall 2024

- Analyzed spectroscopic data of a metal-poor star collected by the Magellan Inamori Kyocera Echelle spectroscope
- Calculated the metallicity, metal abundances, radial velocity, effective temperature, surface gravity, and microturbulence of the star
- Connected quantitative findings to likely history of star

Technical Skills

Languages: Python, C++, Java
Scientific Computing: NumPy, SciPy, AstroPy, Pandas, Matplotlib, Git/GitHub, data reduction, machine learning
Astronomy: Photometry of stars and rocky bodies, Spectroscopy of stars and AGN, AstroImageJ